

# graph-theory-symbol:

A set of  $\text{\LaTeX}$  commands to have symbols designed for  
Graph Theory

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## Abstract

This documentation is for  $\text{\LaTeX}$  package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

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## 1 Introduction

This documentation is for  $\text{\LaTeX}$  package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:

<https://www.gnu.org/licenses/>.

The git repository of this source code is also available at:

<https://github.com/LLyaudet/GraphTheorySymbol/>.

Clone it with this URL:

[git@github.com:LLyaudet/GraphTheorySymbol.git](https://github.com:LLyaudet/GraphTheorySymbol.git).

This is our first  $\text{\LaTeX}$  package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better  $\text{\LaTeX}$  code, please submit a pull request on GitHub.

## 2 Usage

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix “relation”.

$\backslash\text{GTS@newcommand}$  This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.

Arguments:

$\{\langle commandname \rangle\}$  The name of the command to create.

$\{\langle commandcode \rangle\}$  The code of the command to create.

$\backslash\text{GTSsetMainColor}$  This macro will set the main color used in GraphTheorySymbol. This color has initial value “black”.

Argument:

$\{\langle color \rangle\}$  The color that must be used by GraphTheorySymbol afterward.

$\backslash\text{GTSsetSecondColor}$  This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

$\{\langle color \rangle\}$  The color that must be used by GraphTheorySymbol afterward.

$\backslash\text{GTSadjacency}$  This macro will draw an adjacency symbol. A first drawing didn’t include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to “spoon” and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property to remind of the classical textual notation of adjacency relation as  $\text{Adj}(\dots)$ .

$A.\backslash\text{GTSadjacency}/A.$  or  $A.\backslash\text{GTSa}/A.$  will yield  $A\mathbin{\circlearrowleft}A$ .

$u\backslash\text{GTSadjacency}/v$  or  $u\backslash\text{GTSa}/v$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\$u\backslash\text{GTSadjacency}/v\$$  or  $\$u\backslash\text{GTSa}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSadjacencyRelation}$  This macro will draw an adjacency symbol.

$\backslash\text{GTSaR}\ \$u\backslash\text{GTSadjacencyRelation}/v\$$  or  $\$u\backslash\text{GTSaR}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSunadjacency}$  This macro will draw an unadjacency symbol.

$\backslash\text{GTSu}\ A.\backslash\text{GTSunadjacency}/A.$  or  $A.\backslash\text{GTSu}/A.$  will yield  $A\mathbin{\circlearrowright}A$ .

$u\backslash\text{GTSunadjacency}/v$  or  $u\backslash\text{GTSu}/v$  will yield  $u\mathbin{\circlearrowright}v$ .

$\$u\backslash\text{GTSunadjacency}/v\$$  or  $\$u\backslash\text{GTSu}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSunadjacencyRelation}$  This macro will draw an unadjacency symbol.

$\backslash\text{GTSuR}\ \$u\backslash\text{GTSunadjacencyRelation}/v\$$  or  $\$u\backslash\text{GTSuR}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSdirectedAdjacencyUp}$  This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUp}/A.$  or  $A.\backslash\text{GTSdAU}/A.$  will yield  $A\mathbin{\circlearrowleft}A$ .

$u\backslash\text{GTSdirectedAdjacencyUp}/v$  or  $u\backslash\text{GTSdAU}/v$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\$u\backslash\text{GTSdirectedAdjacencyUp}/v\$$  or  $\$u\backslash\text{GTSdAU}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSdirectedAdjacencyUp-Relation}$  This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

$\backslash\text{GTSdAUR}\ \$u\backslash\text{GTSdirectedAdjacencyUpRelation}/v\$$  or  $\$u\backslash\text{GTSdAUR}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSnegatedDirected-AdjacencyUp}$  This macro will draw a negated directed adjacency upward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyUp}/A.$  or  $A.\backslash\text{GTSnDAU}/A.$  will yield  $A\mathbin{\circlearrowright}A$ .

$\backslash\text{GTSnDAU}\ u\backslash\text{GTSnegatedDirectedAdjacencyUp}/v$  or  $u\backslash\text{GTSnDAU}/v$  will yield  $u\mathbin{\circlearrowright}v$ .

$\$u\backslash\text{GTSnegatedDirectedAdjacencyUp}/v\$$  or  $\$u\backslash\text{GTSnDAU}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSnegatedDirected-AdjacencyUpRelation}$  This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

$\backslash\text{GTSnDAUR}\ \$u\backslash\text{GTSdirectedAdjacencyUpRelation}/v\$$  or  $\$u\backslash\text{GTSdAUR}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSdirectedAdjacencyDown}$  This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$A.\backslash\text{GTSdirectedAdjacencyDown}/A.$  or  $A.\backslash\text{GTSdAD}/A.$  will yield  $A\mathbin{\circlearrowright}A$ .

$u\backslash\text{GTSdirectedAdjacencyDown}/v$  or  $u\backslash\text{GTSdAD}/v$  will yield  $u\mathbin{\circlearrowright}v$ .

$\$u\backslash\text{GTSdirectedAdjacencyDown}/v\$$  or  $\$u\backslash\text{GTSdAD}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSdirectedAdjacencyDown-Relation}$  This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

$\backslash\text{GTSdADR}\ \$u\backslash\text{GTSdirectedAdjacencyDownRelation}/v\$$  or  $\$u\backslash\text{GTSdADR}/v\$$  will yield  $u\mathbin{\circlearrowright}v$ .

$\backslash\text{GTSnegatedDirected-AdjacencyDown}$  This macro will draw a negated directed adjacency downward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyDown}/A.$  or  $A.\backslash\text{GTSnDAD}/A.$  will yield  $A\mathbin{\circlearrowleft}A$ .

$\backslash\text{GTSnDAD}\ u\backslash\text{GTSnegatedDirectedAdjacencyDown}/v$  or  $u\backslash\text{GTSnDAD}/v$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\$u\backslash\text{GTSnegatedDirectedAdjacencyDown}/v\$$  or  $\$u\backslash\text{GTSnDAD}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSnegatedDirected-AdjacencyDownRelation}$  This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

$\backslash\text{GTSnDADR}\ \$u\backslash\text{GTSnegatedDirectedAdjacencyDownRelation}/v\$$  or  $\$u\backslash\text{GTSnDADR}/v\$$  will yield  $u\mathbin{\circlearrowleft}v$ .

$\backslash\text{GTSincidence}$  This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$\backslash\text{GTSi}$

$A.\backslash\text{GTSincidence}/A.$  or  $A.\backslash\text{GTSi}/A.$  will yield  $A.\downarrow A.$   
 $v \backslash\text{GTSincidence}/ e$  or  $v \backslash\text{GTSi}/ e$  will yield  $v \downarrow e.$   
 $\$v \backslash\text{GTSincidence}/ e\$$  or  $\$v \backslash\text{GTSi}/ e\$$  will yield  $v\downarrow e.$   
 $\backslash\text{GTSincidenceRelation}$  This macro will draw an incidence symbol with relation spacing in math mode.  
 $\backslash\text{GTSiR}$   $\$v\backslash\text{GTSincidenceRelation}/e\$$  or  $\$v\backslash\text{GTSiR}/e\$$  will yield  $v \downarrow e.$   
 $\backslash\text{GTSnegatedIncidence}$  This macro will draw a negated incidence symbol.  
 $\backslash\text{GTSnI}$   $A.\backslash\text{GTSnegatedIncidence}/A.$  or  $A.\backslash\text{GTSnI}/A.$  will yield  $A.\nmid A.$   
 $v \backslash\text{GTSnegatedIncidence}/ e$  or  $v \backslash\text{GTSnI}/ e$  will yield  $v \nmid e.$   
 $\$v \backslash\text{GTSnegatedIncidence}/ e\$$  or  $\$v \backslash\text{GTSnI}/ e\$$  will yield  $v\nmid e.$   
 $\backslash\text{GTSnegatedIncidence-Relation}$  This macro will draw a negated incidence symbol with relation spacing in math mode.  
 $\$v\backslash\text{GTSnegatedIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSnIR}/e\$$  will yield  $v \nmid e.$   
 $\backslash\text{GTSnIR}$   
 $\backslash\text{GTSoutIncidence}$  This macro will draw a directed incidence upward (out(ward) incidence) symbol. The  
 $\backslash\text{GTSOI}$  operand on the left of the symbol should be considered the vertex to which the arc/the  
operand on the right of the symbol is incident.  
 $A.\backslash\text{GTSoutIncidence}/A.$  or  $A.\backslash\text{GTSOI}/A.$  will yield  $A.\uparrow A.$   
 $v \backslash\text{GTSoutIncidence}/ e$  or  $v \backslash\text{GTSOI}/ e$  will yield  $v \uparrow e.$   
 $\$v \backslash\text{GTSoutIncidence}/ e\$$  or  $\$v \backslash\text{GTSOI}/ e\$$  will yield  $v\uparrow e.$   
 $\backslash\text{GTSoutIncidenceRelation}$  This macro will draw a directed incidence upward (out(ward) incidence) symbol with  
 $\backslash\text{GTSOIR}$  relation spacing in math mode.  
 $\$v\backslash\text{GTSoutIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSOIR}/e\$$  will yield  $v \uparrow e.$   
 $\backslash\text{GTSnegatedOutIncidence}$  This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.  
 $\backslash\text{GTSnOI}$   $A.\backslash\text{GTSnegatedOutIncidence}/A.$  or  $A.\backslash\text{GTSnOI}/A.$  will yield  $A.\nmid A.$   
 $v \backslash\text{GTSnegatedOutIncidence}/ e$  or  $v \backslash\text{GTSnOI}/ e$  will yield  $v \nmid e.$   
 $\$v \backslash\text{GTSnegatedOutIncidence}/ e\$$  or  $\$v \backslash\text{GTSnOI}/ e\$$  will yield  $v\nmid e.$   
 $\backslash\text{GTSnegatedOutIncidence-Relation}$  This macro will draw a negated directed incidence upward (out(ward) incidence)  
symbol with relation spacing in math mode.  
 $\backslash\text{GTSnOIR}$   $\$v\backslash\text{GTSnegatedOutIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSnOIR}/e\$$  will yield  $v \nmid e.$   
 $\backslash\text{GTSinIncidence}$  This macro will draw a directed incidence downward (in(ward) incidence) symbol. The  
 $\backslash\text{GTSiI}$  operand on the left of the symbol should be considered the vertex to which the arc/the  
operand on the right of the symbol is incident.  
 $A.\backslash\text{GTSinIncidence}/A.$  or  $A.\backslash\text{GTSiI}/A.$  will yield  $A.\downarrow A.$   
 $v \backslash\text{GTSinIncidence}/ e$  or  $v \backslash\text{GTSiI}/ e$  will yield  $v \downarrow e.$   
 $\$v \backslash\text{GTSinIncidence}/ e\$$  or  $\$v \backslash\text{GTSiI}/ e\$$  will yield  $v\downarrow e.$   
 $\backslash\text{GTSinIncidenceRelation}$  This macro will draw a directed incidence downward (in(ward) incidence) symbol with  
 $\backslash\text{GTSiIR}$  relation spacing in math mode.  
 $\$v\backslash\text{GTSinIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSiIR}/e\$$  will yield  $v \downarrow e.$   
 $\backslash\text{GTSnegatedInIncidence}$  This macro will draw a negated directed incidence downward (in(ward) incidence) sym-  
 $\backslash\text{GTSnII}$  bol.  
 $A.\backslash\text{GTSnegatedInIncidence}/A.$  or  $A.\backslash\text{GTSnII}/A.$  will yield  $A.\nmid A.$   
 $v \backslash\text{GTSnegatedInIncidence}/ e$  or  $v \backslash\text{GTSnII}/ e$  will yield  $v \nmid e.$   
 $\$v \backslash\text{GTSnegatedInIncidence}/ e\$$  or  $\$v \backslash\text{GTSnII}/ e\$$  will yield  $v\nmid e.$   
 $\backslash\text{GTSnegatedInIncidence-Relation}$  This macro will draw a negated directed incidence downward (in(ward) incidence)  
symbol with relation spacing in math mode.  
 $\backslash\text{GTSnIIR}$   $\$v\backslash\text{GTSnegatedInIncidenceRelation}/e\$$  or  $\$v\backslash\text{GTSnIIR}/e\$$  will yield  $v \nmid e.$   
 $\backslash\text{GTSadjacencyMonochrome}$  This macro will draw a monochrome adjacency symbol.  
 $\backslash\text{GTSaM}$   $A.\backslash\text{GTSadjacencyMonochrome}/A.$  or  $A.\backslash\text{GTSaM}/A.$  will yield  $A.\circ A.$   
 $u \backslash\text{GTSadjacencyMonochrome}/ v$  or  $u \backslash\text{GTSaM}/ v$  will yield  $u \circ v.$   
 $\$u \backslash\text{GTSadjacencyMonochrome}/ v\$$  or  $\$u \backslash\text{GTSaM}/ v\$$  will yield  $u\circ v.$   
 $\backslash\text{GTSadjacencyMonochrome-Relation}$  This macro will draw a monochrome adjacency symbol with relation spacing in math  
mode.  
 $\backslash\text{GTSaMR}$   $\$u\backslash\text{GTSadjacencyMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSaMR}/v\$$  will yield  $u \circ v.$   
 $\backslash\text{GTSunadjacencyMonochrome}$  This macro will draw a monochrome unadjacency symbol.  
 $\backslash\text{GTSuM}$   $A.\backslash\text{GTSunadjacencyMonochrome}/A.$  or  $A.\backslash\text{GTSuM}/A.$  will yield  $A.\circ A.$   
 $u \backslash\text{GTSunadjacencyMonochrome}/ v$  or  $u \backslash\text{GTSuM}/ v$  will yield  $u \circ v.$   
 $\$u \backslash\text{GTSunadjacencyMonochrome}/ v\$$  or  $\$u \backslash\text{GTSuM}/ v\$$  will yield  $u\circ v.$   
 $\backslash\text{GTSunadjacencyMonochrome-Relation}$  This macro will draw a monochrome unadjacency symbol with relation spacing in  
math mode.  
 $\backslash\text{GTSuMR}$   $\$u\backslash\text{GTSunadjacencyMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSuMR}/v\$$  will yield  $u \circ v.$   
 $\backslash\text{GTSdirectedAdjacencyUp-Monochrome}$  This macro will draw a monochrome directed adjacency upward symbol. The operand  
 $\backslash\text{GTSdAUM}$

on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

$A.\backslash\text{GTSdirectedAdjacencyUpMonochrome}/A.$  or  $A.\backslash\text{GTSdAUM}/A.$  will yield  $A.\overset{\circ}{\rightharpoonup}A.$

$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v$  or  $u\ \backslash\text{GTSdAUM}/\ v$  will yield  $u\overset{\circ}{\rightharpoonup}v.$

$\$u\ \backslash\text{GTSdirectedAdjacencyUpMonochrome}/\ v\$$  or  $\$u\ \backslash\text{GTSdAUM}/\ v\$$  will yield  $u\overset{\circ}{\rightharpoonup}v.$

$\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}$  This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

$\backslash\text{GTSdAUMR}$   $\$u\backslash\text{GTSdirectedAdjacencyUpMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSdAUMR}/v\$$  will yield  $u\overset{\circ}{\rightharpoonup}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}$  This macro will draw a monochrome negated directed adjacency upward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/A.$  or  $A.\backslash\text{GTSnDAUM}/A.$  will yield  $A.\overset{\circ}{\nrightarrow}A.$

$\backslash\text{GTSnDAUM}$   $u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v$  or  $u\ \backslash\text{GTSnDAUM}/\ v$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\$u\ \backslash\text{GTSnegatedDirectedAdjacencyUpMonochrome}/\ v\$$  or  $\$u\ \backslash\text{GTSnDAUM}/\ v\$$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}$  This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

$\backslash\text{GTSnDAUMR}$   $\$u\backslash\text{GTSnegatedDirectedAdjacencyUpMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSnDAUMR}/v\$$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\backslash\text{GTSdirectedAdjacencyDownMonochrome}$  This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

$A.\backslash\text{GTSdirectedAdjacencyDownMonochrome}/A.$  or  $A.\backslash\text{GTSdADM}/A.$  will yield  $A.\overset{\circ}{\searrow}A.$

$u\ \backslash\text{GTSdirectedAdjacencyDownMonochrome}/\ v$  or  $u\ \backslash\text{GTSdADM}/\ v$  will yield  $u\overset{\circ}{\searrow}v.$

$\$u\ \backslash\text{GTSdirectedAdjacencyDownMonochrome}/\ v\$$  or  $\$u\ \backslash\text{GTSdADM}/\ v\$$  will yield  $u\overset{\circ}{\searrow}v.$

$\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}$  This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

$\backslash\text{GTSdADMR}$   $\$u\backslash\text{GTSdirectedAdjacencyDownMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSdADMR}/v\$$  will yield  $u\overset{\circ}{\searrow}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}$  This macro will draw a monochrome negated directed adjacency downward symbol.

$A.\backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/A.$  or  $A.\backslash\text{GTSnDADM}/A.$  will yield  $A.\overset{\circ}{\nrightarrow}A.$

$\backslash\text{GTSnDADM}$   $u\ \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/\ v$  or  $u\ \backslash\text{GTSnDADM}/\ v$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\$u\ \backslash\text{GTSnegatedDirectedAdjacencyDownMonochrome}/\ v\$$  or  $\$u\ \backslash\text{GTSnDADM}/\ v\$$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}$  This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

$\backslash\text{GTSnDADMR}$   $\$u\backslash\text{GTSnegatedDirectedAdjacencyDownMonochromeRelation}/v\$$  or  $\$u\backslash\text{GTSnDADMR}/v\$$  will yield  $u\overset{\circ}{\nrightarrow}v.$

$\backslash\text{GTSincidenceMonochrome}$  This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSincidenceMonochrome}/A.$  or  $A.\backslash\text{GTSiM}/A.$  will yield  $A.\downarrow A.$

$v\ \backslash\text{GTSincidenceMonochrome}/\ e$  or  $v\ \backslash\text{GTSiM}/\ e$  will yield  $v\downarrow e.$

$\$v\ \backslash\text{GTSincidenceMonochrome}/\ e\$$  or  $\$v\ \backslash\text{GTSiM}/\ e\$$  will yield  $v\downarrow e.$

$\backslash\text{GTSincidenceMonochromeRelation}$  This macro will draw a monochrome incidence symbol with relation spacing in math mode.  $\$v\backslash\text{GTSincidenceMonochromeRelation}/e\$$  or  $\$v\backslash\text{GTSiMR}/e\$$  will yield  $v\downarrow e.$

$\backslash\text{GTSiMR}$  This macro will draw a monochrome negated incidence symbol.

$\backslash\text{GTSnegatedIncidenceMonochrome}$   $A.\backslash\text{GTSnegatedIncidenceMonochrome}/A.$  or  $A.\backslash\text{GTSnIM}/A.$  will yield  $A.\nrightarrow A.$

$\backslash\text{GTSnIM}$   $v\ \backslash\text{GTSnegatedIncidenceMonochrome}/\ e$  or  $v\ \backslash\text{GTSnIM}/\ e$  will yield  $v\nrightarrow e.$

$\$v\ \backslash\text{GTSnegatedIncidenceMonochrome}/\ e\$$  or  $\$v\ \backslash\text{GTSnIM}/\ e\$$  will yield  $v\nrightarrow e.$

$\backslash\text{GTSnegatedIncidenceMonochromeRelation}$  This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

$\backslash\text{GTSnIMR}$   $\$v\backslash\text{GTSnegatedIncidenceMonochromeRelation}/e\$$  or  $\$v\backslash\text{GTSnIMR}/e\$$  will yield  $v\nrightarrow e.$

$\backslash\text{GTSoutIncidenceMonochrome}$  This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

$A.\backslash\text{GTSoutIncidenceMonochrome}/A.$  or  $A.\backslash\text{GTSoIM}/A.$  will yield  $A.\uparrow A.$

$v\ \backslash\text{GTSoutIncidenceMonochrome}/\ e$  or  $v\ \backslash\text{GTSoIM}/\ e$  will yield  $v\uparrow e.$

$\$v\ \backslash\text{GTSoutIncidenceMonochrome}/\ e\$$  or  $\$v\ \backslash\text{GTSoIM}/\ e\$$  will yield  $v\uparrow e.$

$\backslash\text{GTSoutIncidenceMonochromeRelation}$  This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

$\backslash\text{GTSoIMR}$   $\$v\backslash\text{GTSoutIncidenceMonochromeRelation}/e\$$  or  $\$v\backslash\text{GTSoIMR}/e\$$  will yield  $v\uparrow e.$

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward)  
`Monochrome` incidence) symbol.  
`\GTSnOIM`  $A.\GTSnegatedOutIncidenceMonochrome/A.$  or  $A.\GTSnOIM/A.$  will yield  $A.\text{\textcircled{\tiny A}}$ .  
 $v\ \GTSnegatedOutIncidenceMonochrome/\ e$  or  $v\ \GTSnOIM/\ e$  will yield  $v\ \text{\textcircled{\tiny e}}$ .  
 $\$v\ \GTSnegatedOutIncidenceMonochrome/\ e\$$  or  $\$v\ \GTSnOIM/\ e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward)  
`MonochromeRelation` incidence) symbol with relation spacing in math mode.  
`\GTSnOIMR`  $\$v\GTSnegatedOutIncidenceMonochromeRelation/e\$$  or  $\$v\GTSnOIMR/e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence)  
`\GTSiIM` symbol. The operand on the left of the symbol should be considered the vertex to which  
the arc/the operand on the right of the symbol is incident.  
 $A.\GTSinIncidenceMonochrome/A.$  or  $A.\GTSiIM/A.$  will yield  $A.\text{\textcircled{\tiny A}}$ .  
 $v\ \GTSinIncidenceMonochrome/\ e$  or  $v\ \GTSiIM/\ e$  will yield  $v\ \text{\textcircled{\tiny e}}$ .  
 $\$v\ \GTSinIncidenceMonochrome/\ e\$$  or  $\$v\ \GTSiIM/\ e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

`\GTSinIncidenceMonochrome-` This macro will draw a monochrome directed incidence downward (in(ward) incidence)  
`Relation` symbol with relation spacing in math mode.  
`\GTSiIMR`  $\$v\GTSinIncidenceMonochromeRelation/e\$$  or  $\$v\GTSiIMR/e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)  
`Monochrome` incidence) symbol.  
`\GTSnIIM`  $A.\GTSnegatedInIncidenceMonochrome/A.$  or  $A.\GTSnIIM/A.$  will yield  $A.\text{\textcircled{\tiny A}}$ .  
 $v\ \GTSnegatedInIncidenceMonochrome/\ e$  or  $v\ \GTSnIIM/\ e$  will yield  $v\ \text{\textcircled{\tiny e}}$ .  
 $\$v\ \GTSnegatedInIncidenceMonochrome/\ e\$$  or  $\$v\ \GTSnIIM/\ e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

`\GTSnegatedInIncidence-` This macro will draw a monochrome negated directed incidence downward (in(ward)  
`MonochromeRelation` incidence) symbol with relation spacing in math mode.  
`\GTSnIIMR`  $\$v\GTSnegatedInIncidenceMonochromeRelation/e\$$  or  $\$v\GTSnIIMR/e\$$  will yield  $v\ \text{\textcircled{\tiny e}}$ .

### 3 Implementation

```

1 < *package>
2 \RequirePackage{amsmath}
3 \RequirePackage{amssymb}
4 \RequirePackage{graphicx}
5 \RequirePackage{fdsymbol}
6 \RequirePackage{lua-unicode-math}
7 \RequirePackage{xcolor}
8
9 \def \GTS@mainColor {black}
10 \def \GTS@secondColor {gray}
11 \def \GTS@mainColorBackup {black}
12 \def \GTS@secondColorBackup {gray}
13 \def \GTS@drawSlash {}
14 \def \GTS@drawUpArrow {}
15 \def \GTS@drawDownArrow {}
16
\GTS@newcommand This macro will enforce that the other macros are called with a trailing slash, avoiding
problems with spaces after macro call.
Arguments:
{<commandname>} The name of the command to create.
{<commandcode>} The code of the command to create.
17 \@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
18   \@ifdefinable{#1}{\def#1/{#2}}
19 }}
\GTSsetMainColor This macro will set the main color used in GraphTheorySymbol. This color has initial value
“black”.
Argument:
{<color>} The color that must be used by GraphTheorySymbol afterward.
20 \newcommand{\GTSsetMainColor}[1]{%
21 \edef\GTS@mainColorTemp{#1}%
22 \edef\GTS@mainColorBackup{\GTS@mainColor}%

```

```

23 \edef\GTS@mainColor{\GTS@mainColorTemp}%
24 }

```

**\GTSsetSecondColor** This macro will set the second color used in GraphTheorySymbol. This color has initial value “gray”.

Argument:

$\{color\}$  The color that must be used by GraphTheorySymbol afterward.

```

25 \newcommand{\GTSsetSecondColor}[1]{%
26 \edef\GTS@secondColorTemp{#1}%
27 \edef\GTS@secondColorBackup{\GTS@secondColor}%
28 \edef\GTS@secondColor{\GTS@secondColorTemp}%
29 }

```

**\GTSadjacency** This macro will draw an adjacency symbol.

```

\GTSa 30 \GTS@newcommand{\GTSadjacency}{%
31 \text{\ensuremath{%
32 \raisebox{-0.2ex}{\scalebox{1.5}[1.5]{%
33 %
34 \raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
35 $\mathcolor{\GTS@secondColor}{\circ}$%
36 }}}%
37 \hspace{-0.39ex}%
38 \raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
39 $\mathcolor{\GTS@secondColor}{\bullet}$%
40 }}}%
41 \hspace{-0.12ex}%
42 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
43 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
44 }}}%
45 \hspace{-0.40ex}%
46 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
47 $\mathcolor{\GTS@secondColor}{\circ}$%
48 }}}%
49 \hspace{-0.39ex}%
50 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
51 $\mathcolor{\GTS@secondColor}{\bullet}$%
52 }}}%
53 \hspace{0.27ex}%
54 \if\GTS@drawUpArrow\empty\else%
55 \hspace{-0.65ex}%
56 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
57 \textcolor{\GTS@mainColor}{\textasciicircum}%
58 }}}%
59 \fi%
60 \if\GTS@drawDownArrow\empty\else%
61 \hspace{-0.65ex}%
62 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
63 \textcolor{\GTS@mainColor}{\textasciicircum}%
64 }}}%
65 \fi%
66 \if\GTS@drawSlash\empty\else%
67 \hspace{-0.65ex}%
68 \raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
69 \textcolor{black}{/}%
70 }}}%
71 \fi%
72 %
73 }}}%
74 }}}%
75 }
76
77 \GTS@newcommand{\GTSa}{\GTSadjacency/}

```

**\GTSadjacencyRelation** This macro will draw an adjacency symbol with relation spacing in math mode.

```

\GTSaR 78 \GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSadjacency}}
79 \GTS@newcommand{\GTSaR}{\GTSadjacencyRelation/}

```

`\GTSunadjacency` This macro will draw an unadjacency symbol.

```
\GTSu 80 \GTS@newcommand{\GTSunadjacency}{%
81 \def\GTS@drawSlash{1}%
82 \GTSadjacency/%
83 \def\GTS@drawSlash{}%
84 }
85
86 \GTS@newcommand{\GTSu}{\GTSunadjacency/}
```

`\GTSunadjacencyRelation` This macro will draw an unadjacency symbol with relation spacing in math mode.

```
\GTSuR 87 \GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSunadjacency/}}
88 \GTS@newcommand{\GTSuR}{\GTSunadjacencyRelation/}
```

`\GTSdirectedAdjacencyUp` This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
\GTSdAU 89 \GTS@newcommand{\GTSdirectedAdjacencyUp}{%
90 \def\GTS@drawUpArrow{1}%
91 \GTSadjacency/%
92 \def\GTS@drawUpArrow{}%
93 }
94
95 \GTS@newcommand{\GTSdAU}{\GTSdirectedAdjacencyUp/}
```

`\GTSdirectedAdjacencyUpRelation` This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```
\GTSdAUR 96 \GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{%
97 \mathrel{\GTSdirectedAdjacencyUp/}%
98 }
99 \GTS@newcommand{\GTSdAUR}{\GTSdirectedAdjacencyUpRelation/}
```

`\GTSnegatedDirectedAdjacencyUp` This macro will draw a negated directed adjacency upward symbol.

```
100 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{%
\GTSnDAU 101 \def\GTS@drawSlash{1}%
102 \GTSdirectedAdjacencyUp/%
103 \def\GTS@drawSlash{}%
104 }
105
106 \GTS@newcommand{\GTSnDAU}{\GTSnegatedDirectedAdjacencyUp/}
```

`\GTSnegatedDirectedAdjacencyUpRelation` This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```
\GTSnDAUR 107 \GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
108 \mathrel{\GTSnegatedDirectedAdjacencyUp/}%
109 }
110 \GTS@newcommand{\GTSnDAUR}{\GTSnegatedDirectedAdjacencyDownRelation/}
```

`\GTSdirectedAdjacencyDown` This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
\GTSdAD 111 \GTS@newcommand{\GTSdirectedAdjacencyDown}{%
112 \def\GTS@drawDownArrow{1}%
113 \GTSadjacency/%
114 \def\GTS@drawDownArrow{}%
115 }
116
117 \GTS@newcommand{\GTSdAD}{\GTSdirectedAdjacencyDown/}
```

`\GTSdirectedAdjacencyDownRelation` This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```
\GTSdADR 118 \GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{%
119 \mathrel{\GTSdirectedAdjacencyDown/}%
120 }
121 \GTS@newcommand{\GTSdADR}{\GTSdirectedAdjacencyDownRelation/}
```

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol.

```

AdjacencyDown 122 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{%
\GTSnDAD 123 \def\GTS@drawSlash{1}%
124 \GTSdirectedAdjacencyDown/%
125 \def\GTS@drawSlash{}}%
126 }
127
128 \GTS@newcommand{\GTSnDAD}{\GTSnegatedDirectedAdjacencyDown/}

```

`\GTSnegatedDirected-` This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```

AdjacencyDownRelation 129 \GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\GTSnDADR 130 \mathrel{\GTSnegatedDirectedAdjacencyDown}/%
131 }
132 \GTS@newcommand{\GTSnDADR}{\GTSnegatedDirectedAdjacencyDownRelation/}

```

`\GTSincidence` This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```

\GTSi 133 \GTS@newcommand{\GTSincidence}{%
134 \text{\ensuremath{%
135 \raisebox{0ex}{\scalebox{1.8}[1.8]{%
136 %
137 \raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
138 $\mathcolor{\GTS@secondColor}{\circ}$%
139 }}%
140 \hspace{-0.39ex}%
141 \raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
142 $\mathcolor{\GTS@secondColor}{\bullet}$%
143 }}%
144 \hspace{-0.12ex}%
145 \raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
146 $\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
147 }}%
148 \hspace{-0.18ex}%
149 \raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
150 $\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
151 }}%
152 \hspace{0.2ex}%
153 \if\GTS@drawUpArrow\empty\else%
154 \hspace{-0.65ex}%
155 \raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
156 \textcolor{\GTS@secondColor}{\textasciicircum}}%
157 }}%
158 \fi%
159 \if\GTS@drawDownArrow\empty\else%
160 \hspace{-0.65ex}%
161 \raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
162 \textcolor{\GTS@secondColor}{\textasciicircum}}%
163 }}%
164 \fi%
165 \if\GTS@drawSlash\empty\else%
166 \hspace{-0.65ex}%
167 \raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
168 \textcolor{black}{/}}%
169 }}%
170 \fi%
171 %
172 }}%
173 }}%
174 }
175
176 \GTS@newcommand{\GTSi}{\GTSincidence/}

```

`\GTSincidenceRelation` This macro will draw an incidence symbol with relation spacing in math mode.

```

\GTSiR 177 \GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSincidence/}}
178 \GTS@newcommand{\GTSiR}{\GTSincidenceRelation/}

```

`\GTSnegatedIncidence` This macro will draw a negated incidence symbol.

```

\GTSnI 179 \GTS@newcommand{\GTSnegatedIncidence}{%
180 \def\GTS@drawSlash{1}%
181 \GTSincidence/%
182 \def\GTS@drawSlash{}}%
183 }
184
185 \GTS@newcommand{\GTSnI}{\GTSnegatedIncidence/}

```

`\GTSnegatedIncidence-Relation` This macro will draw a negated incidence symbol with relation spacing in math mode.

```

\GTSnIR 186 \GTS@newcommand{\GTSnegatedIncidenceRelation}{%
187 \mathrel{\GTSnegatedIncidence/}%
188 }
189 \GTS@newcommand{\GTSnIR}{\GTSnegatedIncidenceRelation/}

```

`\GTSoutIncidence` This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSoI 190 \GTS@newcommand{\GTSoutIncidence}{%
191 \def\GTS@drawUpArrow{1}%
192 \GTSincidence/%
193 \def\GTS@drawUpArrow{}}%
194 }
195
196 \GTS@newcommand{\GTSoI}{\GTSoutIncidence/}

```

`\GTSoutIncidenceRelation` This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

\GTSoIR 197 \GTS@newcommand{\GTSoutIncidenceRelation}{%
198 \mathrel{\GTSoutIncidence/}%
199 }
200 \GTS@newcommand{\GTSoIR}{\GTSoutIncidenceRelation/}

```

`\GTSnegatedOutIncidence` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```

\GTSnOI 201 \GTS@newcommand{\GTSnegatedOutIncidence}{%
202 \def\GTS@drawSlash{1}%
203 \GTSoutIncidence/%
204 \def\GTS@drawSlash{}}%
205 }
206
207 \GTS@newcommand{\GTSnOI}{\GTSnegatedOutIncidence/}

```

`\GTSnegatedOutIncidence-Relation` This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

\GTSnOIR 208 \GTS@newcommand{\GTSnegatedOutIncidenceRelation}{%
209 \mathrel{\GTSnegatedOutIncidence/}%
210 }
211 \GTS@newcommand{\GTSnOIR}{\GTSnegatedOutIncidenceRelation/}

```

`\GTSinIncidence` This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSiI 212 \GTS@newcommand{\GTSinIncidence}{%
213 \def\GTS@drawDownArrow{1}%
214 \GTSincidence/%
215 \def\GTS@drawDownArrow{}}%
216 }
217
218 \GTS@newcommand{\GTSiI}{\GTSinIncidence/}

```

`\GTSinIncidenceRelation` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSiIR 219 \GTS@newcommand{\GTSinIncidenceRelation}{%
220 \mathrel{\GTSinIncidence/}%
221 }
222 \GTS@newcommand{\GTSiIR}{\GTSinIncidenceRelation/}

```

`\GTSnegatedInIncidence` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```

\GTSnII 223 \GTS@newcommand{\GTSnegatedInIncidence}{%
224 \def\GTS@drawSlash{1}%
225 \GTSinIncidence/%
226 \def\GTS@drawSlash{}}%
227 }
228
229 \GTS@newcommand{\GTSnII}{\GTSnegatedInIncidence/}

```

`\GTSnegatedInIncidence-` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol  
`Relation` with relation spacing in math mode.

```

\GTSnIIR 230 \GTS@newcommand{\GTSnegatedInIncidenceRelation}{%
231 \mathrel{\GTSnegatedInIncidence/}%
232 }
233 \GTS@newcommand{\GTSnIIR}{\GTSnegatedInIncidenceRelation/}

```

`\GTSadjacencyMonochrome` This macro will draw a monochrome adjacency symbol.

```

\GTSaM 234 \GTS@newcommand{\GTSadjacencyMonochrome}{%
235 \GTSsetMainColor{black}%
236 \GTSsetSecondColor{black}%
237 \GTSadjacency/%
238 \GTSsetMainColor{\GTS@mainColorBackup}%
239 \GTSsetSecondColor{\GTS@secondColorBackup}%
240 }
241
242 \GTS@newcommand{\GTSaM}{\GTSadjacencyMonochrome/}

```

`\GTSadjacencyMonochrome-` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.  
`Relation`

```

\GTSaMR 243 \GTS@newcommand{\GTSadjacencyMonochromeRelation}{%
244 \mathrel{\GTSadjacencyMonochrome/}%
245 }
246 \GTS@newcommand{\GTSaMR}{\GTSadjacencyMonochromeRelation/}

```

`\GTSunadjacencyMonochrome` This macro will draw a monochrome unadjacency symbol.

```

\GTSuM 247 \GTS@newcommand{\GTSunadjacencyMonochrome}{%
248 \def\GTS@drawSlash{1}%
249 \GTSadjacencyMonochrome/%
250 \def\GTS@drawSlash{}}%
251 }
252
253 \GTS@newcommand{\GTSuM}{\GTSunadjacencyMonochrome/}

```

`\GTSunadjacencyMonochrome-` This macro will draw a monochrome unadjacency symbol with relation spacing in math  
`Relation` mode.

```

\GTSuMR 254 \GTS@newcommand{\GTSunadjacencyMonochromeRelation}{%
255 \mathrel{\GTSunadjacencyMonochrome/}%
256 }
257 \GTS@newcommand{\GTSuMR}{\GTSunadjacencyMonochromeRelation/}

```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol. The operand on  
`Monochrome` the left of the symbol should be considered the source of the arc; the operand on the right  
`\GTSdAUM` of the symbol should be considered the target of the arc.

```

258 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
259 \def\GTS@drawUpArrow{1}%
260 \GTSadjacencyMonochrome/%
261 \def\GTS@drawUpArrow{}}%
262 }
263
264 \GTS@newcommand{\GTSdAUM}{\GTSdirectedAdjacencyUpMonochrome/}

```

`\GTSdirectedAdjacencyUp-` This macro will draw a monochrome directed adjacency upward symbol with relation spacing  
`MonochromeRelation` in math mode.

```

\GTSdAUMR 265 \GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
266 \mathrel{\GTSdirectedAdjacencyUpMonochrome/}%
267 }
268 \GTS@newcommand{\GTSdAUMR}{\GTSdirectedAdjacencyUpMonochromeRelation/}

```

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol.

`AdjacencyUpMonochrome` 269 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%`  
`\GTSnDAUM` 270 `\def\GTS@drawSlash{1}%`  
271 `\GTSdirectedAdjacencyUpMonochrome/%`  
272 `\def\GTS@drawSlash{}%`  
273 `}`  
274  
275 `\GTS@newcommand{\GTSnDAUM}{\GTSnegatedDirectedAdjacencyUpMonochrome/}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency upward symbol with relation  
`AdjacencyUpMonochrome-` spacing in math mode.

`Relation` 276 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%`  
`\GTSnDAUMR` 277 `\mathrel{\GTSnegatedDirectedAdjacencyUpMonochrome/}%`  
278 `}`  
279 `\GTS@newcommand{\GTSnDAUMR}{%`  
280 `\GTSnegatedDirectedAdjacencyDownMonochromeRelation/%`  
281 `}`

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol. The operand on  
`Monochrome` the left of the symbol should be considered the target of the arc; the operand on the right  
`\GTSdADM` of the symbol should be considered the source of the arc.

282 `\GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%`  
283 `\def\GTS@drawDownArrow{1}%`  
284 `\GTSadjacencyMonochrome/%`  
285 `\def\GTS@drawDownArrow{}%`  
286 `}`  
287  
288 `\GTS@newcommand{\GTSdADM}{\GTSdirectedAdjacencyDownMonochrome/}`

`\GTSdirectedAdjacencyDown-` This macro will draw a monochrome directed adjacency downward symbol with relation  
`MonochromeRelation` spacing in math mode.

`\GTSdADMR` 289 `\GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%`  
290 `\mathrel{\GTSdirectedAdjacencyDownMonochrome/}%`  
291 `}`  
292 `\GTS@newcommand{\GTSdADMR}{\GTSdirectedAdjacencyDownMonochromeRelation/}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol.

`AdjacencyDownMonochrome` 293 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%`  
`\GTSnDADM` 294 `\def\GTS@drawSlash{1}%`  
295 `\GTSdirectedAdjacencyDownMonochrome/%`  
296 `\def\GTS@drawSlash{}%`  
297 `}`  
298  
299 `\GTS@newcommand{\GTSnDADM}{\GTSnegatedDirectedAdjacencyDownMonochrome/}`

`\GTSnegatedDirected-` This macro will draw a monochrome negated directed adjacency downward symbol with  
`AdjacencyDownMonochrome-` relation spacing in math mode.

`Relation` 300 `\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%`  
`\GTSnDADMR` 301 `\mathrel{\GTSnegatedDirectedAdjacencyDownMonochrome/}%`  
302 `}`  
303 `\GTS@newcommand{\GTSnDADMR}{%`  
304 `\GTSnegatedDirectedAdjacencyDownMonochromeRelation/%`  
305 `}`

`\GTSincidenceMonochrome` This macro will draw a monochrome incidence symbol. The operand on the left of the  
`\GTSiM` symbol should be considered the vertex to which the edge/the operand on the right of the  
symbol is incident.

306 `\GTS@newcommand{\GTSincidenceMonochrome}{%`  
307 `\GTSsetMainColor{black}%`  
308 `\GTSsetSecondColor{black}%`  
309 `\GTSincidence/%`  
310 `\GTSsetMainColor{\GTS@mainColorBackup}%`  
311 `\GTSsetSecondColor{\GTS@secondColorBackup}%`  
312 `}`  
313  
314 `\GTS@newcommand{\GTSiM}{\GTSincidenceMonochrome/}`

`\GTSincidenceMonochrome-` This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```

Relation 315 \GTS@newcommand{\GTSincidenceMonochromeRelation}{%
\GTSiMR 316 \mathrel{\GTSincidenceMonochrome/}%
317 }
318 \GTS@newcommand{\GTSiMR}{\GTSincidenceMonochromeRelation/}

```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol.

```

Monochrome 319 \GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\GTSnIM 320 \def\GTS@drawSlash{1}%
321 \GTSincidenceMonochrome/%
322 \def\GTS@drawSlash{}%
323 }
324
325 \GTS@newcommand{\GTSnIM}{\GTSnegatedIncidenceMonochrome/}

```

`\GTSnegatedIncidence-` This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```

MonochromeRelation 326 \GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{%
\GTSnIMR 327 \mathrel{\GTSnegatedIncidenceMonochrome/}%
328 }
329 \GTS@newcommand{\GTSnIMR}{\GTSnegatedIncidenceMonochromeRelation/}

```

`\GTSoutIncidenceMonochrome` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSoIM 330 \GTS@newcommand{\GTSoutIncidenceMonochrome}{%
331 \def\GTS@drawUpArrow{1}%
332 \GTSincidenceMonochrome/%
333 \def\GTS@drawUpArrow{}%
334 }
335
336 \GTS@newcommand{\GTSoIM}{\GTSoutIncidenceMonochrome/}

```

`\GTSoutIncidenceMonochrome-` This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

Relation 337 \GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{%
\GTSoIMR 338 \mathrel{\GTSoutIncidenceMonochrome/}%
339 }
340 \GTS@newcommand{\GTSoIMR}{\GTSoutIncidenceMonochromeRelation/}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```

Monochrome 341 \GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
\GTSnOIM 342 \def\GTS@drawSlash{1}%
343 \GTSoutIncidenceMonochrome/%
344 \def\GTS@drawSlash{}%
345 }
346
347 \GTS@newcommand{\GTSnOIM}{\GTSnegatedOutIncidenceMonochrome/}

```

`\GTSnegatedOutIncidence-` This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```

MonochromeRelation 348 \GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
\GTSnOIMR 349 \mathrel{\GTSnegatedOutIncidenceMonochrome/}%
350 }
351 \GTS@newcommand{\GTSnOIMR}{\GTSnegatedOutIncidenceMonochromeRelation/}

```

`\GTSinIncidenceMonochrome` This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```

\GTSiIM 352 \GTS@newcommand{\GTSinIncidenceMonochrome}{%
353 \def\GTS@drawDownArrow{1}%
354 \GTSincidenceMonochrome/%
355 \def\GTS@drawDownArrow{}%

```

```

356 }
357
358 \GTS@newcommand{\GTSiIM}{\GTSinIncidenceMonochrome/}

```

**\GTSinIncidenceMonochrome-Relation** This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSiIMR 359 \GTS@newcommand{\GTSinIncidenceMonochromeRelation}{%
360 \mathrel{\GTSinIncidenceMonochrome/}%
361 }
362 \GTS@newcommand{\GTSiIMR}{\GTSinIncidenceMonochromeRelation/}

```

**\GTSnegatedInIncidenceMonochrome** This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```

\GTSnIIM 363 \GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
364 \def\GTS@drawSlash{1}%
365 \GTSinIncidenceMonochrome/%
366 \def\GTS@drawSlash{}%
367 }
368
369 \GTS@newcommand{\GTSnIIM}{\GTSnegatedInIncidenceMonochrome/}

```

**\GTSnegatedInIncidenceMonochromeRelation** This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```

\GTSnIIMR 370 \GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
371 \mathrel{\GTSnegatedInIncidenceMonochrome/}%
372 }
373 \GTS@newcommand{\GTSnIIMR}{\GTSnegatedInIncidenceMonochromeRelation/}

374 \endinput
375 \</package>

```

## 4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

## 5 Change History

v0.1.0 – 2026-08-01

General: Initial version . . . . . 1

v1.0.0 – 2026-08-26

General: First public release . . . . . 1

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